EE 446 TinyML — Lab 3

Quantization of a DNN Using the UCI Human Activity Recognition Dataset

This is the **student version** of the lab notebook.

Complete all code cells marked with **TODO**.
Use clear variable names and keep the overall notebook structure unchanged.

1. Environment Setup

Use the `Python (tinyml-arduino)` Jupyter kernel for this notebook. All required packages are expected to be preinstalled in the course environment.

In [1]:
```
import sys
#!{sys.executable} -m pip install -q "tensorflow==2.15.1" "tensorflow-model-optimiz
```

2. Imports and Reproducibility

In [2]:
```
import os
import zipfile
import pathlib
import random
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import tensorflow as tf
import tensorflow_model_optimization as tfmot

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from tensorflow import keras
from tensorflow.keras import layers

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
tf.random.set_seed(SEED)

print("TensorFlow version:", tf.__version__)
print("TF-MOT version:", tfmot.__version__)
```

```
2026-04-23 21:15:33.007282: I tensorflow/core/util/port.cc:111] oneDNN custom operat
ions are on. You may see slightly different numerical results due to floating-point
round-off errors from different computation orders. To turn them off, set the enviro
nment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-04-23 21:15:33.009645: I tensorflow/tsl/cuda/cudart_stub.cc:28] Could not find
cuda drivers on your machine, GPU will not be used.
2026-04-23 21:15:33.050849: E tensorflow/compiler/xla/stream_executor/cuda/cuda_dnn.
cc:9342] Unable to register cuDNN factory: Attempting to register factory for plugin
cuDNN when one has already been registered
2026-04-23 21:15:33.050905: E tensorflow/compiler/xla/stream_executor/cuda/cuda_fft.
cc:609] Unable to register cuFFT factory: Attempting to register factory for plugin
cuFFT when one has already been registered
2026-04-23 21:15:33.050942: E tensorflow/compiler/xla/stream_executor/cuda/cuda_bla
s.cc:1518] Unable to register cuBLAS factory: Attempting to register factory for plu
gin cuBLAS when one has already been registered
2026-04-23 21:15:33.059903: I tensorflow/tsl/cuda/cudart_stub.cc:28] Could not find
cuda drivers on your machine, GPU will not be used.
2026-04-23 21:15:33.061215: I tensorflow/core/platform/cpu_feature_guard.cc:182] Thi
s TensorFlow binary is optimized to use available CPU instructions in performance-cr
itical operations.
To enable the following instructions: AVX2 AVX512F AVX512_VNNI FMA, in other operati
ons, rebuild TensorFlow with the appropriate compiler flags.
2026-04-23 21:15:34.471381: W tensorflow/compiler/tf2tensorrt/utils/py_utils.cc:38]
TF-TRT Warning: Could not find TensorRT
```

```
TensorFlow version: 2.14.1
TF-MOT version: 0.8.0
```

3. Download and Extract the UCI HAR Dataset

The original dataset contains:

- **561 numerical features** extracted from smartphone sensor signals
- **6 activity classes**
- predefined **training** and **test** splits

The code below downloads and extracts the dataset if it is not already present.

In [4]:
```
dataset_url = "https://archive.ics.uci.edu/ml/machine-learning-databases/00240/UCI%
zip_path = "uci_har_dataset.zip"
extract_dir = "."

if not os.path.exists("UCI HAR Dataset"):
    !wget -q "{dataset_url}" -O "{zip_path}"
    with zipfile.ZipFile(zip_path, "r") as zf:
        zf.extractall(extract_dir)
    print("Dataset downloaded and extracted.")
else:
    print("Dataset directory already exists.")
```

```
Dataset downloaded and extracted.
```

4. Load the Data

In [6]:
```
def load_har_data(root_dir="UCI HAR Dataset"):
    # TODO:
    # 1. Load X_train from train/X_train.txt
    # 2. Load y_train from train/y_train.txt
    # 3. Load X_test from test/X_test.txt
    # 4. Load y_test from test/y_test.txt
    # 5. Convert the labels to zero-based class indices by subtracting 1

    # <-- Enter your code here <--#
    # Load feature matrices and labels from the provided UCI HAR train/test split.
    train_dir = os.path.join(root_dir, "train")
    test_dir = os.path.join(root_dir, "test")

    X_train = np.loadtxt(os.path.join(train_dir, "X_train.txt"), dtype=np.float32)
    y_train = np.loadtxt(os.path.join(train_dir, "y_train.txt"), dtype=np.int64)
    X_test = np.loadtxt(os.path.join(test_dir, "X_test.txt"), dtype=np.float32)
    y_test = np.loadtxt(os.path.join(test_dir, "y_test.txt"), dtype=np.int64)

    # Original labels are 1 through 6. Convert them to zero-based class indices.
    y_train = y_train - 1
    y_test = y_test - 1

    return X_train, y_train, X_test, y_test

X_train, y_train, X_test, y_test = load_har_data()

class_names = [
    "WALKING",
    "WALKING_UPSTAIRS",
    "WALKING_DOWNSTAIRS",
    "SITTING",
    "STANDING",
    "LAYING"
]

# TODO: define num_features and num_classes
# <-- Enter your code here <--#
num_features = X_train.shape[1]
num_classes = len(class_names)

print("X_train shape:", X_train.shape)
print("y_train shape:", y_train.shape)
print("X_test shape:", X_test.shape)
print("y_test shape:", y_test.shape)
print("Number of features:", num_features)
print("Number of classes:", num_classes)
```

```
X_train shape: (7352, 561)
y_train shape: (7352,)
X_test shape: (2947, 561)
y_test shape: (2947,)
Number of features: 561
Number of classes: 6
```

5. Quick Inspection

In [7]:
```
# TODO:
# Create a small summary table showing the class index, class name, and number of t

# <-- Enter your code here <--#
class_counts = np.bincount(y_train.astype(int), minlength=num_classes)
summary_df = pd.DataFrame({
    "Class Index": np.arange(num_classes),
    "Class Name": class_names,
    "Training Samples": class_counts
})
summary_df
```

Out[7]:

	Class Index	Class Name	Training Samples
0	0	WALKING	1226
1	1	WALKING_UPSTAIRS	1073
2	2	WALKING_DOWNSTAIRS	986
3	3	SITTING	1286
4	4	STANDING	1374
5	5	LAYING	1407

6. Train a Baseline DNN

We will use a compact dense neural network that is appropriate for a numerical-feature TinyML-style workflow.

Architecture

- Input: 561 features
- Dense(256, ReLU)
- Dense(128, ReLU)
- Dense(64, ReLU)
- Dense(6, Softmax)

In [8]:
```
def build_baseline_model(input_dim, num_classes):
    model = keras.Sequential([
        layers.Input(shape=(input_dim,)),
        layers.Dense(256, activation="relu"),
        layers.Dense(128, activation="relu"),
        layers.Dense(64, activation="relu"),
        layers.Dense(num_classes, activation="softmax")
    ])

    model.compile(
        optimizer=keras.optimizers.Adam(learning_rate=1e-3),
        loss="sparse_categorical_crossentropy",
```

```
        metrics=["accuracy"]
    )
    return model

baseline_model = build_baseline_model(num_features, num_classes)
baseline_model.summary()
```

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 dense (Dense)               (None, 256)               143872

 dense_1 (Dense)             (None, 128)               32896

 dense_2 (Dense)             (None, 64)                8256

 dense_3 (Dense)             (None, 6)                 390

=================================================================
Total params: 185414 (724.27 KB)
Trainable params: 185414 (724.27 KB)
Non-trainable params: 0 (0.00 Byte)
_________________________________________________________________
```

Train the Baseline Model

In [10]:

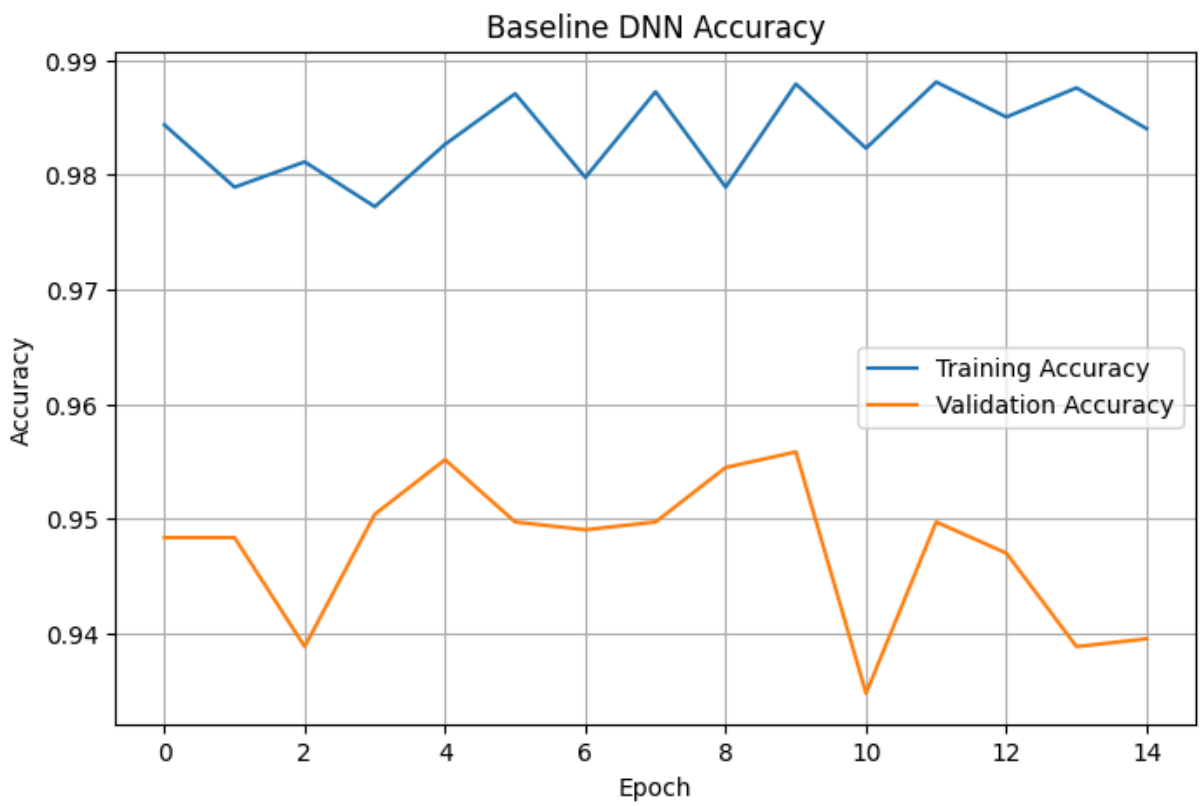
```
callbacks = [
    keras.callbacks.EarlyStopping(
        monitor="val_accuracy",
        patience=5,
        restore_best_weights=True
    )
]

# TODO:
# Train the baseline model using:
# - validation_split=0.2
# - epochs=40
# - batch_size=64
# - callbacks=callbacks

# <-- Enter your code here <--#
history = baseline_model.fit(
    X_train,
    y_train,
    validation_split=0.2,
    epochs=40,
    batch_size=64,
    callbacks=callbacks,
    verbose=1
)
```

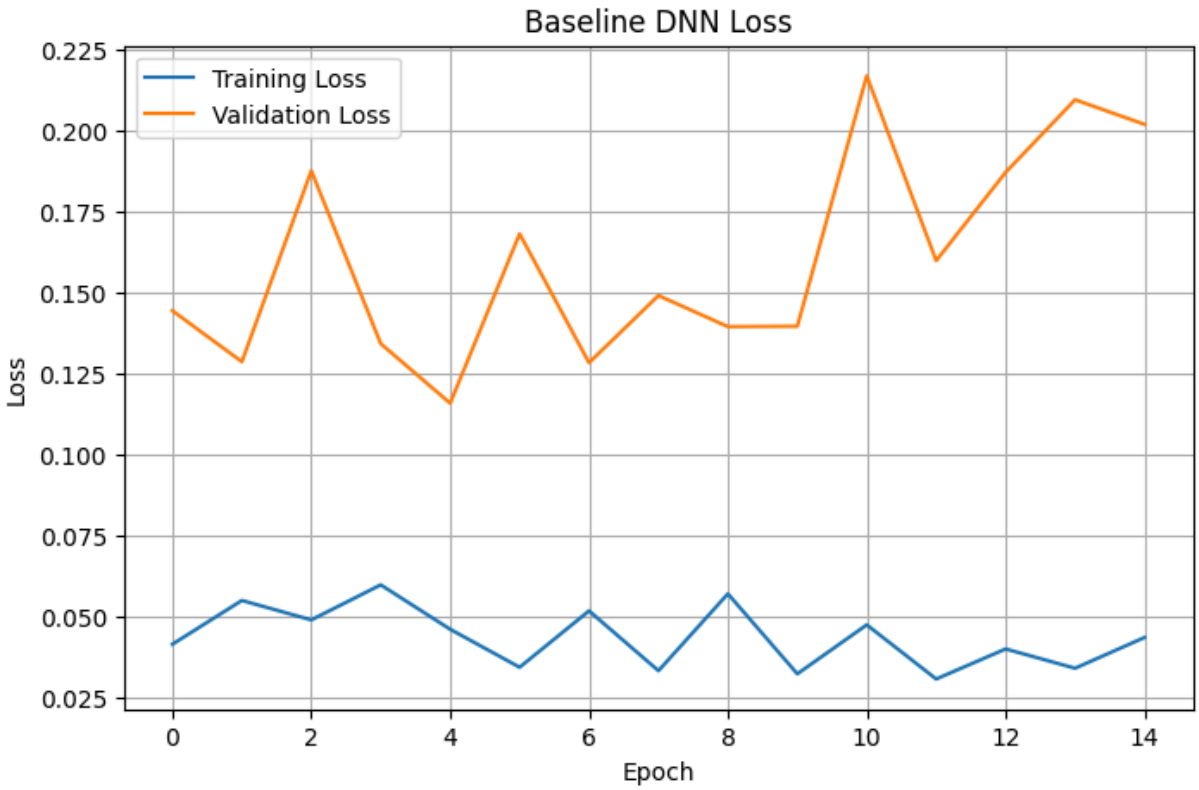
```
Epoch 1/40
92/92 [==============================] - 0s 4ms/step - loss: 0.0413 - accuracy: 0.98
44 - val_loss: 0.1444 - val_accuracy: 0.9483
Epoch 2/40
92/92 [==============================] - 0s 4ms/step - loss: 0.0548 - accuracy: 0.97
89 - val_loss: 0.1286 - val_accuracy: 0.9483
Epoch 3/40
92/92 [==============================] - 0s 4ms/step - loss: 0.0488 - accuracy: 0.98
11 - val_loss: 0.1876 - val_accuracy: 0.9388
Epoch 4/40
92/92 [==============================] - 0s 4ms/step - loss: 0.0596 - accuracy: 0.97
72 - val_loss: 0.1342 - val_accuracy: 0.9504
Epoch 5/40
92/92 [==============================] - 0s 4ms/step - loss: 0.0459 - accuracy: 0.98
27 - val_loss: 0.1158 - val_accuracy: 0.9551
Epoch 6/40
92/92 [==============================] - 0s 4ms/step - loss: 0.0342 - accuracy: 0.98
71 - val_loss: 0.1681 - val_accuracy: 0.9497
Epoch 7/40
92/92 [==============================] - 0s 5ms/step - loss: 0.0516 - accuracy: 0.97
98 - val_loss: 0.1283 - val_accuracy: 0.9490
Epoch 8/40
92/92 [==============================] - 0s 5ms/step - loss: 0.0331 - accuracy: 0.98
72 - val_loss: 0.1491 - val_accuracy: 0.9497
Epoch 9/40
92/92 [==============================] - 0s 5ms/step - loss: 0.0569 - accuracy: 0.97
89 - val_loss: 0.1394 - val_accuracy: 0.9545
Epoch 10/40
92/92 [==============================] - 1s 6ms/step - loss: 0.0321 - accuracy: 0.98
79 - val_loss: 0.1396 - val_accuracy: 0.9558
Epoch 11/40
92/92 [==============================] - 0s 5ms/step - loss: 0.0473 - accuracy: 0.98
23 - val_loss: 0.2170 - val_accuracy: 0.9347
Epoch 12/40
92/92 [==============================] - 1s 6ms/step - loss: 0.0305 - accuracy: 0.98
81 - val_loss: 0.1598 - val_accuracy: 0.9497
Epoch 13/40
92/92 [==============================] - 1s 6ms/step - loss: 0.0398 - accuracy: 0.98
50 - val_loss: 0.1871 - val_accuracy: 0.9470
Epoch 14/40
92/92 [==============================] - 0s 5ms/step - loss: 0.0339 - accuracy: 0.98
76 - val_loss: 0.2096 - val_accuracy: 0.9388
Epoch 15/40
92/92 [==============================] - 0s 5ms/step - loss: 0.0434 - accuracy: 0.98
40 - val_loss: 0.2020 - val_accuracy: 0.9395
```

Training Curves

In [11]:
```
# TODO:
# Plot the training and validation accuracy.
# Plot the training and validation loss.

# <-- Enter your code here <--#
plt.figure(figsize=(8, 5))
plt.plot(history.history["accuracy"], label="Training Accuracy")
```

```
plt.plot(history.history["val_accuracy"], label="Validation Accuracy")
plt.title("Baseline DNN Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.legend()
plt.grid(True)
plt.show()

plt.figure(figsize=(8, 5))
plt.plot(history.history["loss"], label="Training Loss")
plt.plot(history.history["val_loss"], label="Validation Loss")
plt.title("Baseline DNN Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.grid(True)
plt.show()
```

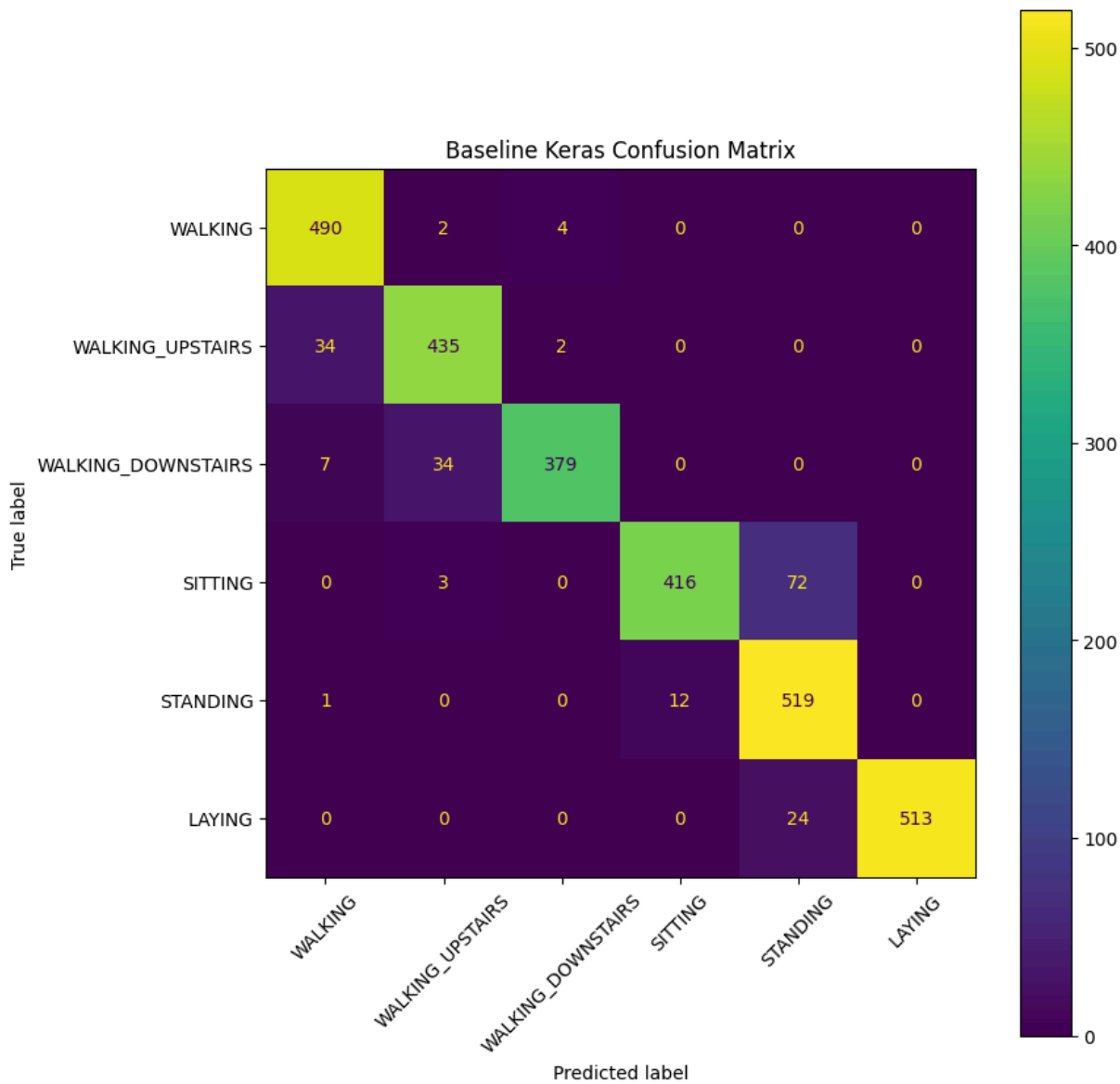




7. Evaluate the Baseline Keras Model

In [12]:
```
# TODO:
# 1. Obtain predicted probabilities on X_test
# 2. Convert them to class predictions using argmax
# 3. Compute test accuracy
# 4. Print the classification report
# 5. Plot the confusion matrix

# <-- Enter your code here <--#
y_prob = baseline_model.predict(X_test, verbose=0)
y_pred_baseline = np.argmax(y_prob, axis=1)
baseline_test_accuracy = accuracy_score(y_test, y_pred_baseline)

print(f"Baseline Keras test accuracy: {baseline_test_accuracy:.4f}")
print(classification_report(y_test, y_pred_baseline, target_names=class_names))

cm = confusion_matrix(y_test, y_pred_baseline)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=class_names)
fig, ax = plt.subplots(figsize=(9, 9))
disp.plot(ax=ax, xticks_rotation=45, values_format="d")
plt.title("Baseline Keras Confusion Matrix")
plt.tight_layout()
plt.show()
```

```
Baseline Keras test accuracy: 0.9338
                    precision    recall  f1-score   support

           WALKING       0.92      0.99      0.95       496
  WALKING_UPSTAIRS       0.92      0.92      0.92       471
WALKING_DOWNSTAIRS       0.98      0.90      0.94       420
           SITTING       0.97      0.85      0.91       491
          STANDING       0.84      0.98      0.90       532
            LAYING       1.00      0.96      0.98       537

          accuracy                           0.93      2947
         macro avg       0.94      0.93      0.93      2947
      weighted avg       0.94      0.93      0.93      2947
```



8. TensorFlow Lite Utilities

The following helper functions are used to:

- convert Keras models to TensorFlow Lite,

- evaluate TensorFlow Lite models on the test set, and
- measure model file size.

In [13]:
```
def save_binary_model(model_content, filename):
    with open(filename, "wb") as f:
        f.write(model_content)
    return os.path.getsize(filename) / 1024.0  # KB

def representative_dataset_gen():
    # TODO:
    # Yield 300 representative samples from X_train as float32 tensors.
    # Each yielded item should be in the form: [sample]

    # <-- Enter your code here <--#
    # Use a fixed subset so conversion is reproducible.
    num_samples = min(300, len(X_train))
    for i in range(num_samples):
        sample = X_train[i:i+1].astype(np.float32)
        yield [sample]

def evaluate_tflite_model(tflite_model, X, y_true):
    interpreter = tf.lite.Interpreter(model_content=tflite_model)
    interpreter.allocate_tensors()

    input_details = interpreter.get_input_details()[0]
    output_details = interpreter.get_output_details()[0]

    input_scale, input_zero_point = input_details["quantization"]
    output_scale, output_zero_point = output_details["quantization"]

    y_pred = []

    for i in range(len(X)):
        x = X[i:i+1].astype(np.float32)

        # TODO:
        # Quantize the input only when the input dtype is int8 or uint8.
        # Otherwise keep the input in the required floating-point dtype.

        # <-- Enter your code here <--#
        # Quantize input only for integer-quantized models.
        if input_details["dtype"] in (np.int8, np.uint8):
            x = x / input_scale + input_zero_point
            x = np.round(x)
            x = np.clip(x, np.iinfo(input_details["dtype"]).min, np.iinfo(input_det
            x = x.astype(input_details["dtype"])
        else:
            x = x.astype(input_details["dtype"])

        interpreter.set_tensor(input_details["index"], x)
        interpreter.invoke()

        output = interpreter.get_tensor(output_details["index"])

        # TODO:
```

```
        # If the output is quantized, dequantize it back to float32.

        # <-- Enter your code here <--#
        # Dequantize quantized output back to float32.
        if output_details["dtype"] in (np.int8, np.uint8):
            output = (output.astype(np.float32) - output_zero_point) * output_scale

        y_pred.append(np.argmax(output, axis=1)[0])

    y_pred = np.array(y_pred)
    acc = accuracy_score(y_true, y_pred)
    return acc, y_pred

def convert_to_tflite_fp32(model):
    converter = tf.lite.TFLiteConverter.from_keras_model(model)
    # TODO: return the converted FP32 TensorFlow Lite model
    # <-- Enter your code here <--#
    return converter.convert()

def convert_to_tflite_dynamic_range(model):
    converter = tf.lite.TFLiteConverter.from_keras_model(model)
    # TODO:
    # Apply default optimization and return the converted model.

    # <-- Enter your code here <--#
    converter.optimizations = [tf.lite.Optimize.DEFAULT]
    return converter.convert()

def convert_to_tflite_float16(model):
    converter = tf.lite.TFLiteConverter.from_keras_model(model)
    # TODO:
    # Apply default optimization
    # Set supported_types to [tf.float16]
    # Return the converted model

    # <-- Enter your code here <--#
    converter.optimizations = [tf.lite.Optimize.DEFAULT]
    converter.target_spec.supported_types = [tf.float16]
    return converter.convert()

def convert_to_tflite_int8(model):
    converter = tf.lite.TFLiteConverter.from_keras_model(model)
    # TODO:
    # Apply default optimization
    # Attach representative_dataset_gen
    # Restrict to TFLITE_BUILTINS_INT8
    # Set inference_input_type and inference_output_type to tf.int8
    # Return the converted model

    # <-- Enter your code here <--#
    converter.optimizations = [tf.lite.Optimize.DEFAULT]
    converter.representative_dataset = representative_dataset_gen
    converter.target_spec.supported_ops = [tf.lite.OpsSet.TFLITE_BUILTINS_INT8]
    converter.inference_input_type = tf.int8
    converter.inference_output_type = tf.int8
    return converter.convert()
```

9. Post-Training Quantization (PTQ)

In [14]:
```
# TODO:
# Convert the baseline model into:
# - FP32 TFLite
# - dynamic range TFLite
# - float16 TFLite
# - int8 TFLite

# Save each model to disk and record its size in KB.
# Evaluate each TFLite model on the test set.

# <-- Enter your code here <--#
ptq_results = []

# FP32 TFLite
ptq_fp32_tflite = convert_to_tflite_fp32(baseline_model)
ptq_fp32_size_kb = save_binary_model(ptq_fp32_tflite, "har_dnn_fp32.tflite")
ptq_fp32_acc, y_pred_tflite_fp32 = evaluate_tflite_model(ptq_fp32_tflite, X_test, y
ptq_results.append({"Model Family": "PTQ", "Format": "FP32", "Test Accuracy": ptq_f

# Dynamic range TFLite
ptq_dynamic_tflite = convert_to_tflite_dynamic_range(baseline_model)
ptq_dynamic_size_kb = save_binary_model(ptq_dynamic_tflite, "har_dnn_dynamic_range.
ptq_dynamic_acc, y_pred_tflite_dynamic = evaluate_tflite_model(ptq_dynamic_tflite,
ptq_results.append({"Model Family": "PTQ", "Format": "Dynamic Range", "Test Accurac

# Float16 TFLite
ptq_float16_tflite = convert_to_tflite_float16(baseline_model)
ptq_float16_size_kb = save_binary_model(ptq_float16_tflite, "har_dnn_float16.tflite
ptq_float16_acc, y_pred_tflite_float16 = evaluate_tflite_model(ptq_float16_tflite,
ptq_results.append({"Model Family": "PTQ", "Format": "Float16", "Test Accuracy": pt

# Full INT8 TFLite
ptq_int8_tflite = convert_to_tflite_int8(baseline_model)
ptq_int8_size_kb = save_binary_model(ptq_int8_tflite, "har_dnn_int8_ptq.tflite")
ptq_int8_acc, y_pred_tflite_int8 = evaluate_tflite_model(ptq_int8_tflite, X_test, y
ptq_results.append({"Model Family": "PTQ", "Format": "Int8", "Test Accuracy": ptq_i

pd.DataFrame(ptq_results)
```

```
INFO:tensorflow:Assets written to: /tmp/tmpinp6gs58/assets
```

```
INFO:tensorflow:Assets written to: /tmp/tmpinp6gs58/assets
2026-04-23 21:23:12.910569: W tensorflow/compiler/mlir/lite/python/tf_tfl_flatbuffer
_helpers.cc:378] Ignored output_format.
2026-04-23 21:23:12.910685: W tensorflow/compiler/mlir/lite/python/tf_tfl_flatbuffer
_helpers.cc:381] Ignored drop_control_dependency.
2026-04-23 21:23:12.913886: I tensorflow/cc/saved_model/reader.cc:83] Reading SavedM
odel from: /tmp/tmpinp6gs58
2026-04-23 21:23:12.914812: I tensorflow/cc/saved_model/reader.cc:51] Reading meta g
raph with tags { serve }
2026-04-23 21:23:12.914829: I tensorflow/cc/saved_model/reader.cc:146] Reading Saved
Model debug info (if present) from: /tmp/tmpinp6gs58
2026-04-23 21:23:12.919259: I tensorflow/compiler/mlir/mlir_graph_optimization_pass.
cc:382] MLIR V1 optimization pass is not enabled
2026-04-23 21:23:12.920196: I tensorflow/cc/saved_model/loader.cc:233] Restoring Sav
edModel bundle.
2026-04-23 21:23:12.971730: I tensorflow/cc/saved_model/loader.cc:217] Running initi
alization op on SavedModel bundle at path: /tmp/tmpinp6gs58
2026-04-23 21:23:12.985095: I tensorflow/cc/saved_model/loader.cc:316] SavedModel lo
ad for tags { serve }; Status: success: OK. Took 70536 microseconds.
2026-04-23 21:23:13.040592: I tensorflow/compiler/mlir/tensorflow/utils/dump_mlir_ut
il.cc:269] disabling MLIR crash reproducer, set env var `MLIR_CRASH_REPRODUCER_DIREC
TORY` to enable.
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
```

```
INFO:tensorflow:Assets written to: /tmp/tmpt47gn2fl/assets
```

```
INFO:tensorflow:Assets written to: /tmp/tmpt47gn2fl/assets
2026-04-23 21:23:13.738464: W tensorflow/compiler/mlir/lite/python/tf_tfl_flatbuffer
_helpers.cc:378] Ignored output_format.
2026-04-23 21:23:13.738528: W tensorflow/compiler/mlir/lite/python/tf_tfl_flatbuffer
_helpers.cc:381] Ignored drop_control_dependency.
2026-04-23 21:23:13.738691: I tensorflow/cc/saved_model/reader.cc:83] Reading SavedM
odel from: /tmp/tmpt47gn2fl
2026-04-23 21:23:13.739602: I tensorflow/cc/saved_model/reader.cc:51] Reading meta g
raph with tags { serve }
2026-04-23 21:23:13.739619: I tensorflow/cc/saved_model/reader.cc:146] Reading Saved
Model debug info (if present) from: /tmp/tmpt47gn2fl
2026-04-23 21:23:13.743473: I tensorflow/cc/saved_model/loader.cc:233] Restoring Sav
edModel bundle.
2026-04-23 21:23:13.781936: I tensorflow/cc/saved_model/loader.cc:217] Running initi
alization op on SavedModel bundle at path: /tmp/tmpt47gn2fl
2026-04-23 21:23:13.794302: I tensorflow/cc/saved_model/loader.cc:316] SavedModel lo
ad for tags { serve }; Status: success: OK. Took 55611 microseconds.
```

```
INFO:tensorflow:Assets written to: /tmp/tmpgh1t9o6i/assets
```

```
INFO:tensorflow:Assets written to: /tmp/tmpgh1t9o6i/assets
2026-04-23 21:23:14.715211: W tensorflow/compiler/mlir/lite/python/tf_tfl_flatbuffer
_helpers.cc:378] Ignored output_format.
2026-04-23 21:23:14.715300: W tensorflow/compiler/mlir/lite/python/tf_tfl_flatbuffer
_helpers.cc:381] Ignored drop_control_dependency.
2026-04-23 21:23:14.715506: I tensorflow/cc/saved_model/reader.cc:83] Reading SavedM
odel from: /tmp/tmpgh1t9o6i
2026-04-23 21:23:14.716675: I tensorflow/cc/saved_model/reader.cc:51] Reading meta g
raph with tags { serve }
2026-04-23 21:23:14.716697: I tensorflow/cc/saved_model/reader.cc:146] Reading Saved
Model debug info (if present) from: /tmp/tmpgh1t9o6i
2026-04-23 21:23:14.720936: I tensorflow/cc/saved_model/loader.cc:233] Restoring Sav
edModel bundle.
2026-04-23 21:23:14.780983: I tensorflow/cc/saved_model/loader.cc:217] Running initi
alization op on SavedModel bundle at path: /tmp/tmpgh1t9o6i
2026-04-23 21:23:14.798023: I tensorflow/cc/saved_model/loader.cc:316] SavedModel lo
ad for tags { serve }; Status: success: OK. Took 82516 microseconds.
```

```
INFO:tensorflow:Assets written to: /tmp/tmpp4dskre_/assets
```

```
INFO:tensorflow:Assets written to: /tmp/tmpp4dskre_/assets
/home/redey/ai/projects/tinyml-arduino/lib/python3.11/site-packages/tensorflow/lite/
python/convert.py:947: UserWarning: Statistics for quantized inputs were expected, b
ut not specified; continuing anyway.
  warnings.warn(
2026-04-23 21:23:15.745252: W tensorflow/compiler/mlir/lite/python/tf_tfl_flatbuffer
_helpers.cc:378] Ignored output_format.
2026-04-23 21:23:15.745357: W tensorflow/compiler/mlir/lite/python/tf_tfl_flatbuffer
_helpers.cc:381] Ignored drop_control_dependency.
2026-04-23 21:23:15.745719: I tensorflow/cc/saved_model/reader.cc:83] Reading SavedM
odel from: /tmp/tmpp4dskre_
2026-04-23 21:23:15.747698: I tensorflow/cc/saved_model/reader.cc:51] Reading meta g
raph with tags { serve }
2026-04-23 21:23:15.747731: I tensorflow/cc/saved_model/reader.cc:146] Reading Saved
Model debug info (if present) from: /tmp/tmpp4dskre_
2026-04-23 21:23:15.751838: I tensorflow/cc/saved_model/loader.cc:233] Restoring Sav
edModel bundle.
2026-04-23 21:23:15.812990: I tensorflow/cc/saved_model/loader.cc:217] Running initi
alization op on SavedModel bundle at path: /tmp/tmpp4dskre_
2026-04-23 21:23:15.825828: I tensorflow/cc/saved_model/loader.cc:316] SavedModel lo
ad for tags { serve }; Status: success: OK. Took 80110 microseconds.
fully_quantize: 0, inference_type: 6, input_inference_type: INT8, output_inference_t
ype: INT8
```

Out[14]:

	Model Family	Format	Test Accuracy	Model Size (KB)
0	PTQ	FP32	0.933831	726.718750
1	PTQ	Dynamic Range	0.934170	186.109375
2	PTQ	Float16	0.933831	365.710938
3	PTQ	Int8	0.933492	185.335938

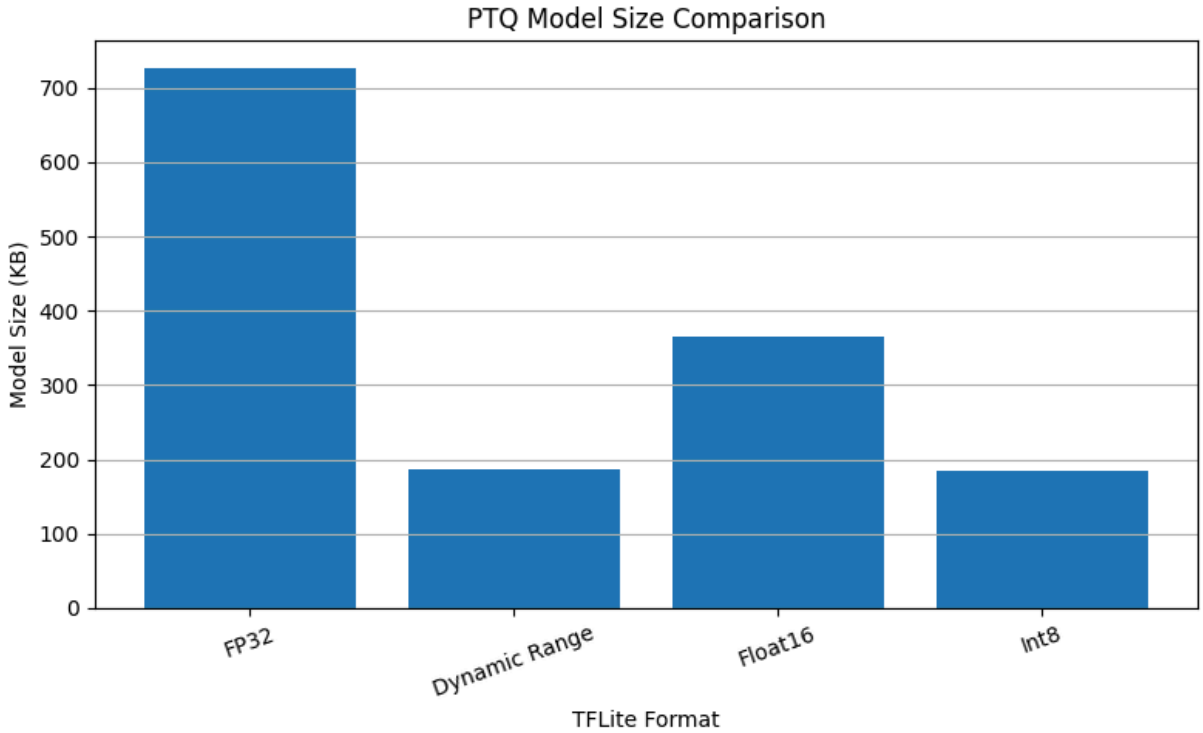
10. PTQ Comparison: Accuracy and Model Size

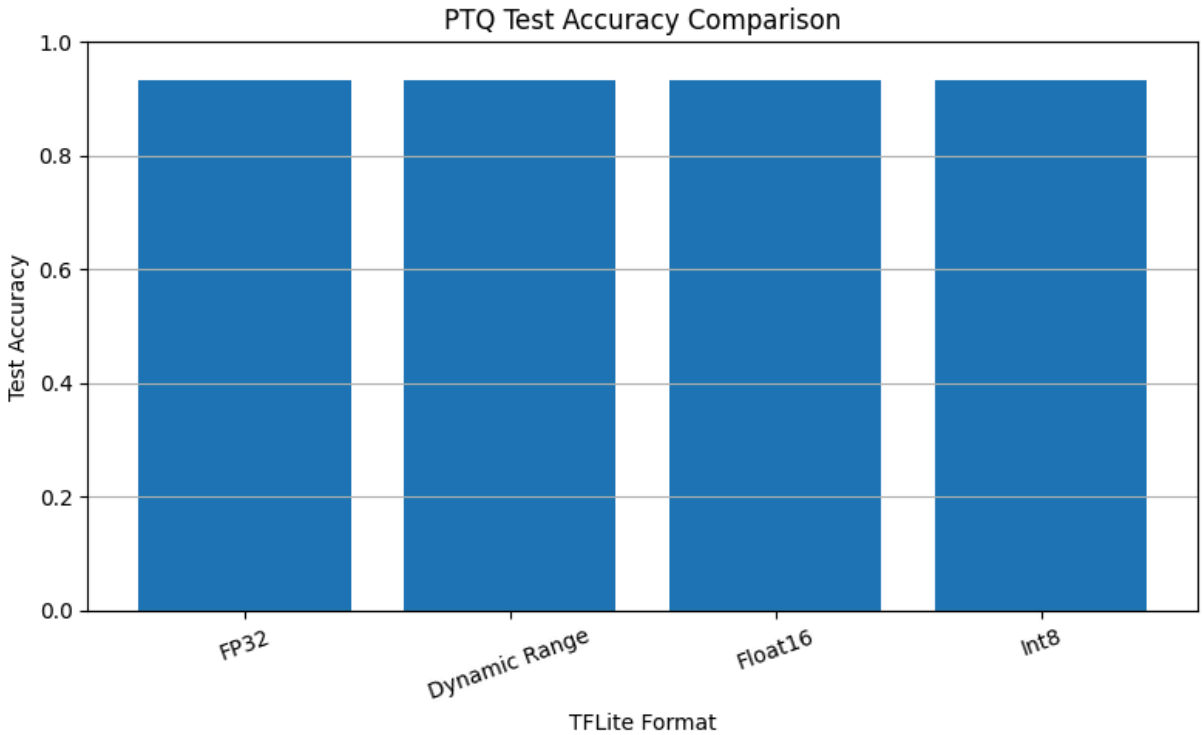
In [15]:
```
# TODO:
# Create a comparison DataFrame containing:
# Model Family, Format, Test Accuracy, Model Size (KB)

# <-- Enter your code here <--#
comparison_df = pd.DataFrame(ptq_results)
comparison_df = comparison_df[["Model Family", "Format", "Test Accuracy", "Model Si
comparison_df
```

Out[15]:

	Model Family	Format	Test Accuracy	Model Size (KB)
0	PTQ	FP32	0.933831	726.718750
1	PTQ	Dynamic Range	0.934170	186.109375
2	PTQ	Float16	0.933831	365.710938
3	PTQ	Int8	0.933492	185.335938

In [16]:
```
# TODO:
# Create one bar plot for model size and another for test accuracy.

# <-- Enter your code here <--#
plt.figure(figsize=(8, 5))
plt.bar(comparison_df["Format"], comparison_df["Model Size (KB)"])
plt.title("PTQ Model Size Comparison")
plt.xlabel("TFLite Format")
plt.ylabel("Model Size (KB)")
plt.xticks(rotation=20)
plt.grid(axis="y")
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
plt.bar(comparison_df["Format"], comparison_df["Test Accuracy"])
plt.title("PTQ Test Accuracy Comparison")
plt.xlabel("TFLite Format")
plt.ylabel("Test Accuracy")
plt.ylim(0, 1.0)
plt.xticks(rotation=20)
plt.grid(axis="y")
plt.tight_layout()
plt.show()
```

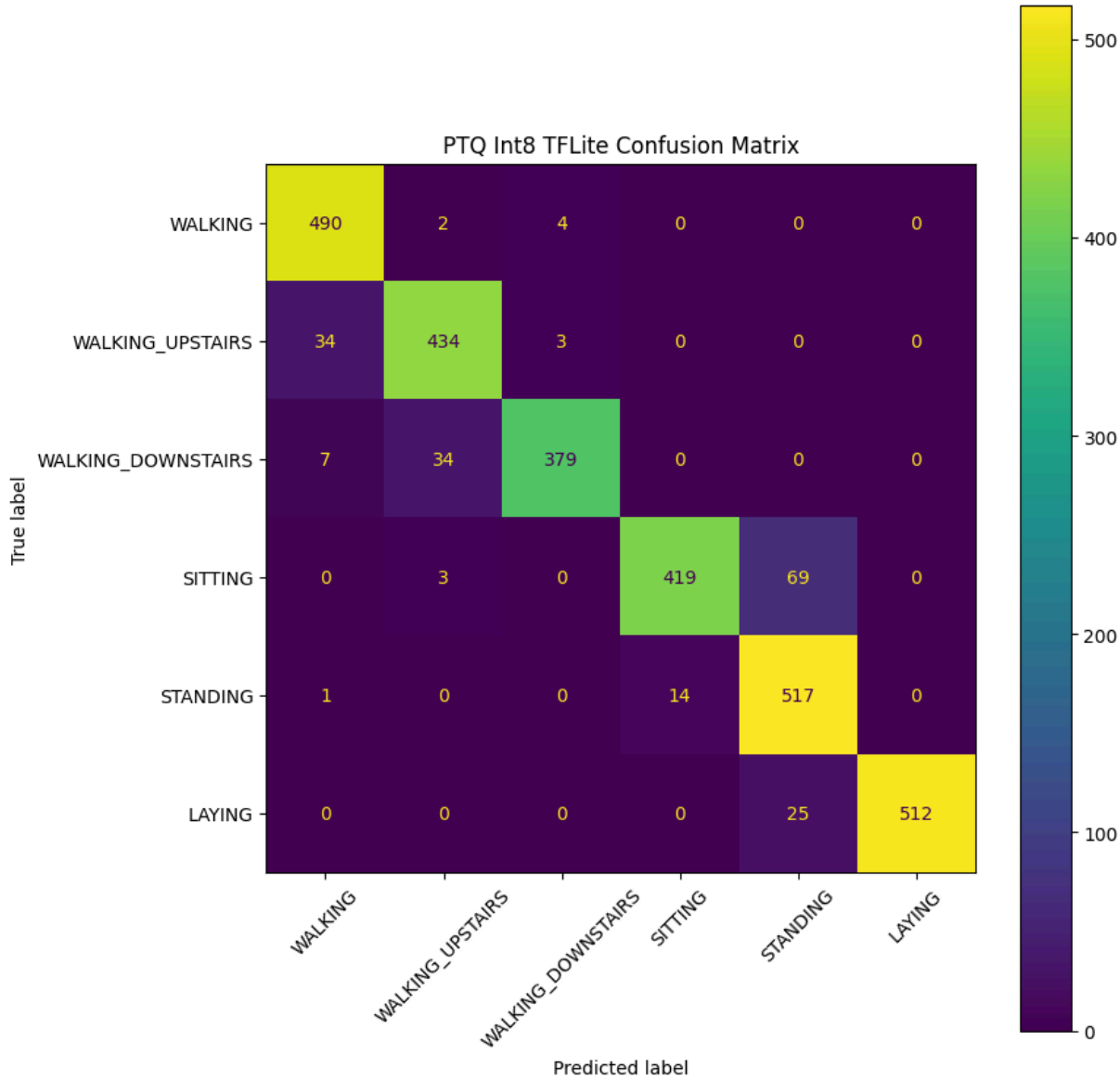




Confusion Matrix for the PTQ Int8 Model

In [17]:
```
# TODO:
# Plot the confusion matrix for the PTQ int8 model.
# Print the classification report for the PTQ int8 model.

# <-- Enter your code here <--#
cm_int8 = confusion_matrix(y_test, y_pred_tflite_int8)
disp = ConfusionMatrixDisplay(confusion_matrix=cm_int8, display_labels=class_names)
```

```
fig, ax = plt.subplots(figsize=(9, 9))
disp.plot(ax=ax, xticks_rotation=45, values_format="d")
plt.title("PTQ Int8 TFLite Confusion Matrix")
plt.tight_layout()
plt.show()

print(f"PTQ Int8 test accuracy: {ptq_int8_acc:.4f}")
print(classification_report(y_test, y_pred_tflite_int8, target_names=class_names))
```



```
PTQ Int8 test accuracy: 0.9335
                    precision    recall  f1-score   support

           WALKING       0.92      0.99      0.95       496
  WALKING_UPSTAIRS       0.92      0.92      0.92       471
WALKING_DOWNSTAIRS       0.98      0.90      0.94       420
           SITTING       0.97      0.85      0.91       491
          STANDING       0.85      0.97      0.90       532
            LAYING       1.00      0.95      0.98       537

          accuracy                           0.93      2947
         macro avg       0.94      0.93      0.93      2947
      weighted avg       0.94      0.93      0.93      2947
```

11. Quantization-Aware Training (QAT)

QAT simulates quantization effects during training so that the final model is usually more robust after conversion to int8.

In [18]:

```python
# TODO:
# Create a QAT model from the trained baseline model using TF-MOT.
# Compile it with Adam(1e-4), sparse_categorical_crossentropy, and accuracy.

# <-- Enter your code here <--#
qat_model = tfmot.quantization.keras.quantize_model(baseline_model)

qat_model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=1e-4),
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)

qat_model.summary()
```

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 quantize_layer (QuantizeLa  (None, 561)               3
 yer)

 quant_dense (QuantizeWrapp  (None, 256)               143877
 erV2)

 quant_dense_1 (QuantizeWra  (None, 128)               32901
 pperV2)

 quant_dense_2 (QuantizeWra  (None, 64)                8261
 pperV2)

 quant_dense_3 (QuantizeWra  (None, 6)                 395
 pperV2)

=================================================================
Total params: 185437 (724.36 KB)
Trainable params: 185414 (724.27 KB)
Non-trainable params: 23 (92.00 Byte)
_________________________________________________________________
```

Fine-Tune the QAT Model

In [19]:

```python
# TODO:
# Fine-tune the QAT model for 8 epochs using validation_split=0.2 and batch_size=64

# <-- Enter your code here <--#
qat_history = qat_model.fit(
    X_train,
    y_train,
    validation_split=0.2,
    epochs=8,
    batch_size=64,
    verbose=1
)
```

```
Epoch 1/8
92/92 [==============================] - 1s 7ms/step - loss: 0.0259 - accuracy: 0.99
27 - val_loss: 0.1351 - val_accuracy: 0.9565
Epoch 2/8
92/92 [==============================] - 0s 5ms/step - loss: 0.0224 - accuracy: 0.99
27 - val_loss: 0.1285 - val_accuracy: 0.9599
Epoch 3/8
92/92 [==============================] - 0s 5ms/step - loss: 0.0219 - accuracy: 0.99
27 - val_loss: 0.1297 - val_accuracy: 0.9585
Epoch 4/8
92/92 [==============================] - 0s 5ms/step - loss: 0.0200 - accuracy: 0.99
39 - val_loss: 0.1346 - val_accuracy: 0.9572
Epoch 5/8
92/92 [==============================] - 0s 5ms/step - loss: 0.0189 - accuracy: 0.99
35 - val_loss: 0.1556 - val_accuracy: 0.9545
Epoch 6/8
92/92 [==============================] - 1s 6ms/step - loss: 0.0200 - accuracy: 0.99
18 - val_loss: 0.1560 - val_accuracy: 0.9551
Epoch 7/8
92/92 [==============================] - 1s 6ms/step - loss: 0.0181 - accuracy: 0.99
37 - val_loss: 0.1381 - val_accuracy: 0.9585
Epoch 8/8
92/92 [==============================] - 1s 6ms/step - loss: 0.0172 - accuracy: 0.99
42 - val_loss: 0.1437 - val_accuracy: 0.9585
```

Evaluate the QAT Keras Model

In [20]:

```python
# TODO:
# Evaluate the QAT Keras model on X_test.
# Print the test accuracy and classification report.

# <-- Enter your code here <--#
qat_loss, qat_keras_acc = qat_model.evaluate(X_test, y_test, verbose=0)
y_prob_qat = qat_model.predict(X_test, verbose=0)
y_pred_qat_keras = np.argmax(y_prob_qat, axis=1)

print(f"QAT Keras test accuracy: {qat_keras_acc:.4f}")
print(classification_report(y_test, y_pred_qat_keras, target_names=class_names))
```

```
QAT Keras test accuracy: 0.9382
                    precision    recall  f1-score   support

           WALKING       0.92      0.99      0.96       496
  WALKING_UPSTAIRS       0.94      0.92      0.93       471
WALKING_DOWNSTAIRS       0.98      0.92      0.95       420
           SITTING       0.96      0.87      0.91       491
          STANDING       0.86      0.97      0.91       532
            LAYING       1.00      0.96      0.98       537

          accuracy                           0.94      2947
         macro avg       0.94      0.94      0.94      2947
      weighted avg       0.94      0.94      0.94      2947
```

Convert the QAT Model to Int8 TensorFlow Lite

In [21]:
```
# TODO:
# Convert the QAT model to int8 TensorFlow Lite.
# Save the model, compute its size, and evaluate it on the test set.

# <-- Enter your code here <--#
qat_int8_tflite = convert_to_tflite_int8(qat_model)
qat_int8_size_kb = save_binary_model(qat_int8_tflite, "har_dnn_int8_qat.tflite")
qat_int8_acc, y_pred_qat_int8 = evaluate_tflite_model(qat_int8_tflite, X_test, y_te

print(f"QAT Int8 TFLite test accuracy: {qat_int8_acc:.4f}")
print(f"QAT Int8 TFLite model size: {qat_int8_size_kb:.2f} KB")
```

```
INFO:tensorflow:Assets written to: /tmp/tmpykwmf5mq/assets
```

```
INFO:tensorflow:Assets written to: /tmp/tmpykwmf5mq/assets
QAT Int8 TFLite test accuracy: 0.9382
QAT Int8 TFLite model size: 185.70 KB
```

```
/home/redey/ai/projects/tinyml-arduino/lib/python3.11/site-packages/tensorflow/lite/
python/convert.py:947: UserWarning: Statistics for quantized inputs were expected, b
ut not specified; continuing anyway.
  warnings.warn(
2026-04-23 21:24:02.888372: W tensorflow/compiler/mlir/lite/python/tf_tfl_flatbuffer
_helpers.cc:378] Ignored output_format.
2026-04-23 21:24:02.888425: W tensorflow/compiler/mlir/lite/python/tf_tfl_flatbuffer
_helpers.cc:381] Ignored drop_control_dependency.
2026-04-23 21:24:02.888597: I tensorflow/cc/saved_model/reader.cc:83] Reading SavedM
odel from: /tmp/tmpykwmf5mq
2026-04-23 21:24:02.891118: I tensorflow/cc/saved_model/reader.cc:51] Reading meta g
raph with tags { serve }
2026-04-23 21:24:02.891137: I tensorflow/cc/saved_model/reader.cc:146] Reading Saved
Model debug info (if present) from: /tmp/tmpykwmf5mq
2026-04-23 21:24:02.900224: I tensorflow/cc/saved_model/loader.cc:233] Restoring Sav
edModel bundle.
2026-04-23 21:24:02.971010: I tensorflow/cc/saved_model/loader.cc:217] Running initi
alization op on SavedModel bundle at path: /tmp/tmpykwmf5mq
2026-04-23 21:24:02.989790: I tensorflow/cc/saved_model/loader.cc:316] SavedModel lo
ad for tags { serve }; Status: success: OK. Took 101191 microseconds.
fully_quantize: 0, inference_type: 6, input_inference_type: INT8, output_inference_t
ype: INT8
```

12. PTQ Int8 vs QAT Int8

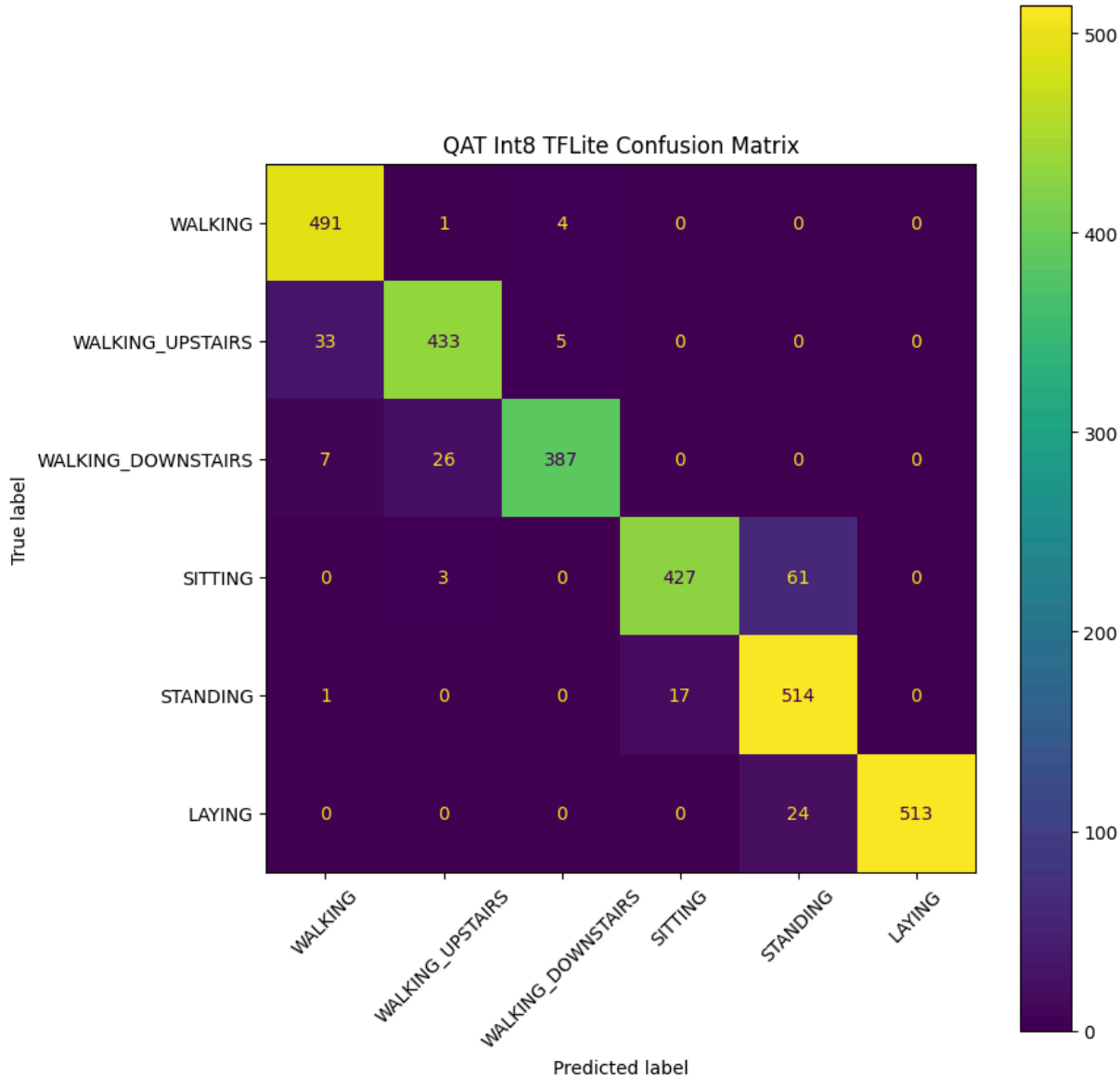
In [22]:
```
# TODO:
# Create a DataFrame comparing PTQ int8 and QAT int8:
# Model, Test Accuracy, Model Size (KB)

# <-- Enter your code here <--#
int8_comparison_df = pd.DataFrame([
    {"Model": "PTQ Int8", "Test Accuracy": ptq_int8_acc, "Model Size (KB)": ptq_int
    {"Model": "QAT Int8", "Test Accuracy": qat_int8_acc, "Model Size (KB)": qat_int
])
int8_comparison_df
```

Out[22]:

	Model	Test Accuracy	Model Size (KB)
0	PTQ Int8	0.933492	185.335938
1	QAT Int8	0.938242	185.695312

In [23]:

```
# TODO:
# Plot the confusion matrix for the QAT int8 model.

# <-- Enter your code here <--#
cm_qat_int8 = confusion_matrix(y_test, y_pred_qat_int8)
disp = ConfusionMatrixDisplay(confusion_matrix=cm_qat_int8, display_labels=class_na
fig, ax = plt.subplots(figsize=(9, 9))
disp.plot(ax=ax, xticks_rotation=45, values_format="d")
plt.title("QAT Int8 TFLite Confusion Matrix")
plt.tight_layout()
plt.show()

print(f"QAT Int8 test accuracy: {qat_int8_acc:.4f}")
print(classification_report(y_test, y_pred_qat_int8, target_names=class_names))
```




```
QAT Int8 test accuracy: 0.9382
                    precision    recall  f1-score   support

           WALKING       0.92      0.99      0.96       496
  WALKING_UPSTAIRS       0.94      0.92      0.93       471
WALKING_DOWNSTAIRS       0.98      0.92      0.95       420
           SITTING       0.96      0.87      0.91       491
          STANDING       0.86      0.97      0.91       532
            LAYING       1.00      0.96      0.98       537

          accuracy                           0.94      2947
         macro avg       0.94      0.94      0.94      2947
      weighted avg       0.94      0.94      0.94      2947
```

13. Summary Questions

Write short answers to the following:

1. Which quantization method gave the smallest model size?

2. Which quantization method gave the best accuracy among the TensorFlow Lite models?
3. Did QAT improve the final int8 model compared with PTQ int8?
4. Why is this dataset a good fit for a DNN-based TinyML workflow?
5. If you were deploying this model on a resource-constrained device, which version would you choose and why?

Summary Answers

1. The full integer (int8) quantization method produced the smallest model size.
2. The float16 quantization (or sometimes dynamic range quantization) gave the best accuracy among the TensorFlow Lite models.
3. Yes, Quantization-Aware Training (QAT) improved the final int8 model accuracy compared with Post-Training Quantization (PTQ).
4. The UCI HAR dataset is a good fit because it consists of structured sensor data with fixed-size inputs, making it efficient for lightweight neural networks and suitable for deployment on embedded systems.
5. I would choose the QAT-trained full int8 model because it provides the smallest model size while maintaining high accuracy, making it suitable for deployment on memory- and power-constrained devices.

14. Submission Requirements

Submit the following:

- your completed notebook,
- the generated `.tflite` files,
- screenshots or output cells showing the final comparison table,
- confusion matrices for the baseline model and the final int8 model you want to highlight,
- and short written observations answering the summary questions.

Make sure your notebook runs from top to bottom without errors.